

Xtrata: The Permanent Object Layer — Revolutionary Use Cases and the Ideas Most Likely to Turn Heads

A use-case and launch-strategy report based on deep research

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Executive Summary

Xtrata is a fully on-chain object platform on Stacks, anchored to Bitcoin, in which media, ownership, and the relationships between objects all live in smart-contract state — turning files into permanent, ownable, machine-readable assets rather than revocable links.

The timing window is open. Four waves converge in 2026: the on-chain music platforms are dead (Sound.xyz offline since January; Nina Protocol goes dark July 22), institutional deletion has turned deliberate (8,000+ US government pages purged), deepfake fraud has made provenance a balance-sheet problem (\$1.5B+ in nine months; EU AI Act rules effective August 2), and AI agents have identity and payment rails with nowhere permanent to publish.

This report develops twelve use cases across five groups — music and culture; truth and evidence; AI, agents, and provenance; games and play; and moonshots. Four stand out: the undeletable album dropped into the Nina shutdown discourse; a monthly cadence of single-artifact rescues that makes permanence recurring press; a war-crime evidence vault usable by witnesses who have never touched crypto, via sponsored zero-STX claims; and contract-enforced remix lineage that pays every ancestor automatically.

The honest constraints, in one breath: Arweave undercuts storage costs 5–100×, engineering is one founder plus AI assistants, the SDK is unpublished, and the audience is 117 X followers with zero press.

First moves are cheap and fast: weeks 0–4 fix the developer-agent surface (npm release, llms.txt, MCP server, extended agent skill); month 1 fires the Nina-timed music drop; month 2 rescues a deleted civic dataset; month 3 pilots the evidence vault with an NGO; months 3–9 build the remix-royalty contract with XTRATA FM as its venue. The window closes when someone else ships first.

1. What Xtrata Actually Is — and Why It Is Technically Different

The “so what” up front: Xtrata is not another NFT platform and not another storage network. It is a working system in which the media itself, the ownership record, and the relationships between objects all live in one smart-contract layer — verified on-chain, not promised in a whitepaper. That machine is the prerequisite for Chapter 3,

because the use cases exploit *relationships between permanent objects*, not permanence alone.

1.1 The Core Machine: Fully On-Chain Objects on Stacks, Anchored to Bitcoin

1.1.1 Media lives inside inscriptions: 16,384-byte chunks, SIP-009 NFTs, sealed immutable, SHA-256 chain-hash content addressing with native dedupe; staged resumable uploads; 32 MiB cap; live contract xtrata-v3-2-3 with ~2,800 v3 inscriptions

An Xtrata inscription is a SIP-009 NFT (the Stacks equivalent of ERC-721) whose content is stored *inside the contract*, not behind a link. Files are split into fixed 16,384-byte chunks written into contract storage, then “sealed,” after which the content is cryptographically immutable¹. Addressing is content-based: an incremental SHA-256 chain hash over the ordered chunks produces a unique fingerprint, and the `begin-or-get mint` function doubles as deduplication — re-uploading an identical payload returns the existing token instead of minting a duplicate². Uploads are staged and resumable: a session survives interruption for roughly 30 days (4,320 blocks), so multi-megabyte files do not depend on a single transaction succeeding². The hard ceiling today is 2,048 chunks, i.e., 32 MiB per object¹.

This is not vaporware. The canonical contract, `xtrata-v3-2-3`, deployed 2026-06-08 as the third generation of a lineage that began in January 2026, held 2,807 inscriptions at last read — objects ranging from images and text to a 4.5 MB MP3 and a 697 KB self-contained HTML application^{3 4}.

1.1.2 The object graph: parent→child ownership links (parent escrowed at mint), dependency references (existence-only, up to 50), on-chain reply threads, recursive HTML via `/i/{id}` — relationships are smart-contract-legible, unlike Bitcoin Ordinals where graph data is invisible to script

The decisive feature is not storage but the **object graph** — three typed relationship primitives stored in contract state. *Parent→child links* encode ownership: to mint a child, you must own the parent, which the contract escrows during the mint and returns with the new inscription; multi-parent mints are supported. *Dependencies* are existence-only references — anything may reference up to 50 other inscriptions without owning them — enabling modular media such as an HTML player that loads

¹ Xtrata Docs (stxtrata/xtrata, main-staging).

<https://github.com/stxtrata/xtrata/blob/main-staging/xtrata-1.0/docs/xtrata-inscription-handbook.md>

² Xtrata Docs (stxtrata/xtrata, main-staging).

https://github.com/stxtrata/xtrata/blob/main-staging/xtrata-1.0/XTRATA_AGENT_SKILL.md

³ Hiro mainnet API.

<https://api.mainnet.hiro.so/extended/v1/address/SP3JNSEXAZP4BDSHV0DN3M8R3P0MY0EEBQQZX743X/transactions>

⁴ Xtrata. <https://xtrata.xyz/i/312>

its audio and cover art from separate tokens. *Reply threads* let anyone attach an on-chain response to any inscription⁵. Recursive HTML composes these through same-origin `/i/{id}` URLs that serve reconstructed bytes with CORS-open, range-streamable responses — so an entire application can be assembled from on-chain parts^{5 1}.

Why this matters: Bitcoin Ordinals has native parent/child provenance (the parent's UTXO is spent into the child's mint), but that link is legible only to off-chain indexers — Bitcoin script cannot read it, which is why one on-chain Ordinals game needed an estimated 2,000,000 indexer requests to compute state for a single 10k collection^{6 7}. Xtrata's graph is readable and enforceable by Clarity smart contracts at the settlement layer itself. Relationships stop being metadata conventions and become machine-actionable facts.

1.2 The UX Breakthrough: Sponsored Transactions and Free Drops

1.2.1 Zero-STX claims and creates; drops contract live since 2026-07 with 32 campaigns; why this inverts crypto onboarding

Permanence technology has always demanded crypto-native users. Stacks supports protocol-native sponsored transactions — a third party can pay a user's network fee⁸ — and Xtrata builds product on top of it. The drops contract (`xtrata-drops-v1-0`, live 2026-07-11) runs sponsored free-claim campaigns: the creator prepays fees, claimers need zero STX; 32 campaigns ran in its first ten days⁹. Sponsored market listings work the same way for buyers. This inverts the onboarding funnel: institutions, labels, or DAOs can pay for permanence *on behalf of* people who have never touched crypto — the difference between a crypto toy and public infrastructure.

1.3 The Open Surface: Market, SDK, XTRATA FM, and Agent Documentation

1.3.1 Multi-currency market (STX/sBTC/USDCx), escrow vaults, arcade-scores with secp256k1 attestations, XTRATA FM streaming multi-MB MP3s from chain, 1,106-line agent skill targeting aibtc agents

The surrounding stack is unusually complete for a project this young. Fixed-price escrow markets settle in STX, sBTC (Bitcoin on Stacks), or USDCx (Circle-issued USDC on Stacks); sBTC vault contracts gate premium access; arcade leaderboard contracts

⁵ Xtrata Docs (`stxtrata/xtrata`, main-staging).

<https://github.com/stxtrata/xtrata/blob/main-staging/xtrata-1.0/docs/recursive-inscriptions.md>

⁶ Ordinal Theory Handbook. <https://docs.ordinals.com/inscriptions/provenance.html>

⁷ `ordinals/ord` GitHub issue #3719. <https://github.com/ordinals/ord/issues/3719>

⁸ Stacks Documentation. <https://docs.stacks.co/concepts/sbtc/auxiliary-features/fee-sponsorship>

⁹ Hiro mainnet API.

<https://api.mainnet.hiro.so/v2/contracts/interface/SP3JNSEXAZP4BDSHV0DN3M8R3P0MY0EEBQQZX743X/xtrata-drops-v1-0>

accept secp256k1-signed score attestations, so inscriptions double as playable games with verifiable high scores¹⁰. XTRATA FM — a persistent radio widget with no playlist server — streams multi-megabyte MP3s straight from chain data via HTTP range requests; one verified track reconstructs 4,533,058 bytes from 277 on-chain chunks^{11 4}. Two SDK packages ship with 15+ quickstarts, including deterministic reconstruction with hash verification¹². Most unusually, a 1,106-line agent skill document teaches autonomous AI agents — explicitly aibtc-platform agents — to mint, dedupe, and trade inscriptions under deny-mode transaction guardrails². The surface is open to machines, not just humans.

1.4 Honest Traction Snapshot

1.4.1 Early-stage reality: ~3,209 lifetime inscriptions mostly team mints, 7 lifetime market listings, zero press, 117 X followers — the ideas in this report are the growth strategy

Credibility requires the downside numbers. Across all four contract versions Xtrata has roughly 3,209 lifetime inscriptions, and creator sampling shows most were minted by team or insider wallets^{3 13}. The market has seven lifetime listings (six STX, one sBTC, zero USDCx) with no visible secondary sales; the market page currently reads “no live listings right now”^{14 15}. Press coverage is effectively zero, the official X account has 117 followers, and there is no Discord, Telegram, token, or fundraiser¹⁶. The machine works; the audience does not yet exist. That gap is not a footnote — the ideas that follow are the proposed growth strategy.

1.5 Differentiation vs Ordinals, Arweave, IPFS, and Conventional NFTs

1.5.1 Table: cost per MB, permanence model, programmability, relationship semantics across Xtrata / Bitcoin Ordinals / Ethereum NFT+IPFS / Arweave / Filecoin-Sia-Storj — Xtrata wins on semantics + sponsored UX + Bitcoin finality at Stacks cost; Arweave cheaper per raw MB

Cost figures are fee-sample-derived estimates anchored to 2026-07-21 network conditions; treat them as order-of-magnitude, not quotes¹⁷.

¹⁰ Xtrata Docs (stxtrata/xtrata, main-staging).

<https://github.com/stxtrata/xtrata/blob/main-staging/xtrata-1.0/docs/contract-inventory.md>

¹¹ Xtrata. <https://xtrata.xyz/assets/main-BuSmoc1z.js>

¹² Xtrata Docs (stxtrata/xtrata, main-staging).

<https://github.com/stxtrata/xtrata/blob/main-staging/xtrata-1.0/docs/sdk/README.md>

¹³ Xtrata Xplorer + Hiro mainnet API. <https://xtrata.xyz/xplorer>

¹⁴ Xtrata Market + Hiro mainnet API. <https://xtrata.xyz/market>

¹⁵ Gamma.io.

<https://stacks.gamma.io/collections/SP3JNSEXAZP4BDSHV0DN3M8R3P0MY0EEBQQZX743X.xtrata-v2-1-0/270>

¹⁶ X (Twitter). <https://x.com/XtrataLayers>

¹⁷ mempool.space API + Hiro mainnet API. <https://mempool.space/api/v1/fees/recommended>

Platform	Cost per MB (est., 2026-07)	Permanence model	Programmability / contract-legibility	Relationship semantics	Sponsored / free UX
Xtrata (Stacks)	~\$0.1–2 ¹⁷	Stacks chain state, Bitcoin finality post-Nakamoto ¹⁸	Clarity contracts read and enforce the graph	Multi-parent escrow, ≤50 dependencies, replies — all on-chain	Native: zero-STX drops and sponsored listings ⁹
Bitcoin Ordinals (recursive)	\$160–315 today; \$3.9k–15.8k in busy regimes; \$6.6k measured ^{17 19}	Bitcoin L1 itself — the strongest anchor	None: UTXO/PSBT only; script cannot read graph data	Parent/child exists (tag 3) but indexer-legible only; no dependencies or replies ⁶	Impossible: no sponsored fees on L1
Ethereum NFT + IPFS	Media off-chain; ~10% of NFT media fully on-chain; 3.91% of IPFS images vanished within 6 months in one study ²⁰	Pinning/server ops; lapses ⇒ link rot	Token contract programmable; media is an off-chain URI pointer	Per-contract convention; no media graph	Meta-transaction relayers only (fragile, non-native)
Arweave	~\$0.019 one-time ²¹	~200-year storage endowment (economic projection, not guarantee) ²²	No native ownership, market, or object layer	File tags only	None
Filecoin / Sia / Storj	~\$0.0002–0.001/GB/month and below ²³	Time-limited deals/contracts (Filecoin ~540 days; Sia ~90 days; Storj deletes on non-payment);	None — bytes-as-a-service	None	None

¹⁸ Stacks Documentation. <https://docs.stacks.co/learn/block-production/what-was-the-nakamoto-upgrade>

¹⁹ The Defiant. <https://thedefiant.io/news/nfts-and-web3/quantum-cats-ordinals-collection-pays-homage-to-satoshi>

²⁰ Decrypt. <https://decrypt.co/138676/are-your-nfts-safe-how-to-protect-digital-assets-from-disaster>

²¹ kkdemian (Arweave network dashboard, citing ViewBlock). <https://kkdemian.com/blog/arweave-ar-permanent-storage-endowment-demand-risk>

²² ArDrive (Arweave ecosystem). <https://ardrive.io/what-is-arweave>

²³ SEC (BitFuFu 20-F correspondence) / Datarecovery.com / Storj Docs. <https://www.sec.gov/Archives/edgar/data/1296774/000110465923110394/filename1.htm>

Platform	Cost per MB (est., 2026-07)	Permanence model	Programmability / contract- legibility	Relationship semantics	Sponsored / free UX
		lapse $\Rightarrow$ loss ²³			

The table resolves to a clear strategic position. Xtrata loses on raw dollars per megabyte: Arweave stores a megabyte for roughly two cents against Xtrata’s estimated \$0.1–2, so bulk archiving belongs elsewhere^{17 21}. It wins on everything that makes an object an *asset*. Bitcoin Ordinals offers the strongest anchor but prices media at \$160–15,800 per megabyte, caps standard transactions near 100 KB, and hides its relationship data from any program^{17 6}. Ethereum NFTs delegate media to servers and pinning services that demonstrably fail — over \$160 million in transaction value is tied to disappeared assets in one analysis²⁰. Filecoin, Sia, and Storj rent bytes on renewable contracts with no concept of an owned object at all²³. Xtrata is the only row combining on-chain media, Bitcoin finality at ~5-second confirmations¹⁸, a contract-enforceable relationship graph, and sponsored zero-fee flows in one layer. One honest caveat: a 2026 Stacks proposal on chain-state pruning could push very old state toward archive nodes, so the “rebuild from chain alone” guarantee should be tracked as it evolves²⁴.

The mental model for the rest of this report: Xtrata sells *meaningful permanence* — permanent bytes whose ownership, lineage, and dependencies are machine-readable. Chapter 2 shows why the demand for it peaks right now; Chapter 3 shows what that unlocks.

2. The Timing Window: Four Converging Waves

The timing case for Xtrata rests on dates and dollar figures, not sentiment. Four external waves converged in the eighteen months to July 2026: every platform that promised music permanence has collapsed, deletion of public information has gone institutional, deepfake fraud has turned content authenticity into a balance-sheet problem, and AI agents have acquired money and identity with nowhere permanent to publish.

Wave	Trigger events	What died	What opens for Xtrata
2.1 On-chain music graveyard	Sound.xyz offline Jan 16, 2026 ²⁵ ; Nina Protocol shuts Jul 22,	The “permanence” platforms — their media was never on-	Music objects that outlive any operator

²⁴ Stacks Forum. <https://forum.stacks.org/t/chain-state-pruning-and-at-block-proposed-change/18685>

²⁵ Sound.xyz. <https://sound.xyz>

Wave	Trigger events	What died	What opens for Xtrata
	2026	chain	
2.2 The Great Deletion	25% of 2013–23 web pages gone ²⁶ ; 8,000+ US government pages purged	The assumption that institutions preserve	An un-deletable public record
2.3 Provenance panic	\$1.5B+ deepfake-fraud losses in nine months ²⁷	Metadata-based authenticity, stripped in transit	An immutable anchor of record
2.4 Agent commerce	HTTP-402 agent payments in sBTC; ERC-8004 identity live ²⁸	The excuse that agents lack rails	Immutable publication — the missing third leg

The pattern across all four rows is identical: the artifact dies, or the proof dies, the moment it depends on a revocable institution — a startup’s servers, a government website, a social platform’s transcoder. Each wave carries primary-source numbers, and each peaks inside the same eighteen-month window. That simultaneity is the strategy. The audiences primed to care — artists, archivists, newsrooms, standards bodies, agent builders — are all seeking permanence at once, and none of the incumbents they once trusted can supply it today.

2.1 The On-Chain Music Platform Graveyard

2.1.1 Sound.xyz (\$25M, a16z) offline 2026-01-16 with media never on-chain; Nina Protocol shutting down 2026-07-22; Catalog pinned media on IPFS — the exact failure Xtrata prevents, and the narrative vacuum is open now

Sound.xyz raised some \$25 million, including a \$20 million Series A led by Andreessen Horowitz, having paid \$5.5 million to its first 500 invited artists²⁹. On January 16, 2026 it went offline; the farewell note insists “the music and metadata are stored in decentralized storage” — true of the tokens, false of the product, because the audio files and player were never on-chain²⁵. Nina Protocol reached roughly 40,000 monthly users and 20,000 releases³⁰, announced a phased shutdown on May 28, 2026, and goes offline tomorrow, July 22³¹ — founded explicitly to save music from moments like

²⁶ ISSN (summarizing Pew Research Center, “When Online Content Disappears”). https://www.issn.org/newsletter_issn/when-online-content-disappears/

²⁷ Veriff. <https://www.veriff.com/fraud/deepfake-fraud-cost-2026>

²⁸ AIBTC. <https://aibtc.com/llms.txt>

²⁹ Billboard. <https://www.billboard.com/pro/sound-xyz-nft-platform-20m-andreessen-horowitz-snoop-dogg/>

³⁰ Crypto Briefing. <https://cryptobriefing.com/solana-music-launch-spotify-challenger/>

³¹ Nina Protocol. <https://www.ninaprotocol.com>

“MySpace’s servers going down”³². Catalog’s one-of-one records persist only while the company pays to pin them on IPFS (file hosting that survives only while someone keeps paying)³³; Royal closed in late 2024 after raising ~\$71 million³⁴. Every flagship permanence play kept the music off-chain and died with its website — the exact failure Xtrata’s architecture prevents — leaving the “permanent home for music” narrative unclaimed.

2.2 The Great Deletion: Link Rot and Institutional Data Purges

2.2.1 Pew 2024: 25% of 2013–2023 pages gone; 2025 US gov purges (8,000+ pages, 2,000+ datasets, CDC/climate tools); YouTube’s deletion of 120–150k Syrian war-crime evidence videos

Pew Research found 25% of webpages existing between 2013 and 2023 already inaccessible — 38% for 2013 — and 21% of government pages carrying a broken link²⁶. Then deletion turned deliberate: more than 8,000 pages vanished from over a dozen US federal websites in early 2025, spanning vaccines, veterans’ care, hate crimes, and scientific research³⁵, while 2,000+ datasets disappeared from data.gov and federal climate portals went dark³⁶. A federal judge vacated the removals as “arbitrary and capricious,” yet restoration stayed partial³⁷. YouTube’s classifiers had already erased an estimated 120,000–150,000 Syrian war-crime evidence videos³⁸, and the Internet Archive, the web’s supposed fallback, went offline and read-only in October 2024 after a 31-million-account breach³⁹. Preservation outsourced to goodwill is preservation already lost.

2.3 The Provenance Panic

³² Resident Advisor. <https://ra.co/news/80797>

³³ Decentral. <https://www.decentral.io/articles/do-music-nfts-still-matter-my-journey-with-catalog-to-find-out>

³⁴ Chartlex. <https://www.chartlex.com/blog/business/music-nft-web3-post-mortem-2026>

³⁵ Society of Environmental Journalists (citing New York Times analysis). <https://www.sej.org/headlines/thousands-us-government-web-pages-have-been-taken-down-friday>

³⁶ Freie Universität Berlin, Earth Sciences Library. <https://www.geo.fu-berlin.de/en/bibliotheken/Aktuelles/Verschwindende-Daten---Trump-Administration-entfernt-Klimainformationen-von-Regierungswebsites.html>

³⁷ Fierce Healthcare. <https://www.fiercehealthcare.com/regulatory/judge-vacates-trump-administrations-removal-health-web-pages>

³⁸ WITNESS (written submission to the UN Special Rapporteur on freedom of expression, via OHCHR, citing Syrian Archive). <https://www.ohchr.org/Documents/Issues/Opinion/ContentRegulation/Witness.pdf>

³⁹ The Verge. <https://www.theverge.com/2024/10/14/24269741/internet-archive-online-read-only-data-breach-outage>

2.3.1 Deepfake fraud ~\$1.5B+ in 9 months of 2025; C2PA manifests structurally strippable in transit — demand for an immutable anchor of record

Reported deepfake-fraud losses topped \$1.5 billion in the first nine months of 2025 alone²⁷; human detection of high-quality deepfakes runs at roughly 24.5% accuracy — worse than a coin flip⁴⁰. The industry’s answer is C2PA, a standard embedding a signed provenance manifest inside the media file; adoption is real: 6,000+ member organizations and camera/phone signing. But the manifest is metadata, and platforms strip metadata during upload and transcoding, C2PA included — the largest gap between signing infrastructure and verification reality in any major technical standard, per one 2026 analysis⁴¹. The EU AI Act’s machine-readable marking rule takes effect August 2, 2026 — twelve days after this report⁴². Provenance that cannot survive a screenshot needs an anchor outside the file.

2.4 The Agent Commerce Wave

2.4.1 aibtc live with HTTP-402 agent payments in sBTC; ERC-8004-on-Stacks registries mainnet-deployed; agents need identity, payment rails, and immutable publication — Stacks has the first two, Xtrata is the third

On Stacks, the agent economy is already operational. aibtc runs a live network where AI agents hold self-custodial wallets and pay each other 100 satoshis of sBTC per message via HTTP-402, the web’s dormant “payment required” code repurposed as a payment rail²⁸. ERC-8004, the standard giving each agent an on-chain identity NFT and a reputation registry, launched on mainnet January 29, 2026, drawing roughly 83,000 registered agents across 18 networks within months⁴³; its identity, reputation, and validation registries are mainnet-deployed on Stacks⁴⁴, and aibtc requires an on-chain agent identity for its trading competition⁴⁵. The wider x402 ecosystem logged about 165 million agent transactions by April 2026, though genuine commerce remains small — roughly \$28,000 per day⁴⁶. Agents on Bitcoin’s rails already have identity and money; the missing third leg is immutable publication, and Xtrata’s agent skill file already teaches aibtc agents to inscribe autonomously⁴⁷.

⁴⁰ Adaptive Security. <https://www.adaptivesecurity.com/blog/ai-deepfake-trends-the-complete-2025-2026-guide-to-statistics-threats-detection-and-defense-strategies>

⁴¹ AIBuzz. <https://aibuzz.blog/ai-watermarking-vs-metadata-vs-fingerprinting/>

⁴² Truescreen. <https://truescreen.io/articles/c2pa-standard-history-limitations/>

⁴³ CryptoSlate. <https://cryptoslate.com/ethereum-aims-to-stop-rogue-ai-agents-from-stealing-trust-with-new-erc-8004-but-can-it-really/>

⁴⁴ aibtcdev (GitHub). <https://github.com/aibtcdev/erc-8004-stacks>

⁴⁵ AIBTC. <https://aibtc.com/docs/identity.txt>

⁴⁶ Presenc AI. <https://presenc.ai/research/x402-protocol-adoption-tracker-2026>

⁴⁷ Xtrata (GitHub, stxtrata/xtrata, OPTIMISATIONS branch). https://github.com/stxtrata/xtrata/blob/OPTIMISATIONS/xtrata-1.0/XTRATA_AGENT_SKILL.md

Chapter 3 maps each wave to use cases: un-deletable music objects and on-chain remix lineage (2.1), civic-data rescue and war-crime evidence preservation (2.2), on-chain manifests of record and verifiably human work (2.3), and permanent agent portfolios with machine-readable publication (2.4). The window is open because all four audiences are looking at once; it closes the moment someone ships first.

3. The Head-Turning Use Cases

Every use case in this chapter passed a single filter: the pain must be documented with hard evidence, and the cure must require something only Xtrata does — a smart-contract-legible object graph on permanent media, sponsored zero-STX distribution, or media that is itself chain data. Eight are buildable on contracts live on mainnet today — 2,807 inscriptions on the current v3 contract, a sponsored-drops contract deployed July 11, 2026, escrow markets settling in STX, sBTC, and USDCx^{48 49} — and each of the other four names its one missing piece.

#	Use case	Sector	Decisive Xtrata primitive	Feasibility
UC1	The album that outlives the label	Music	Chunked immutable media + sponsored drops	Now
UC2	Contract-enforced remix lineage	Music	Parent escrow at mint	Near — royalty-split contract
UC3	The unshutable radio station	Music	XTRATA FM over recursive /i/{id} bytes	Now
UC4	War-crime evidence vault	Truth	Sponsored claims + hash-native integrity	Now — needs NGO partner
UC5	Civic data rescue	Truth	Queryable objects + reply threads	Now
UC6	Self-correcting scientific record	Truth	Inseparable reply-thread retractions	Near — journal pilot
UC7	Certified-human registry	AI	Wallet-signed immutable publication	Now

⁴⁸ Hiro Mainnet API. <https://api.mainnet.hiro.so/v2/contracts/call-read/SP3JNSEXAZP4BDSHV0DN3M8R3P0MY0EEBQQZX743X/xtrata-v3-2-3/{get-fee-unit,get-minted-count,get-last-token-id}>

⁴⁹ Xtrata. <https://xtrata.xyz>

#	Use case	Sector	Decisive Xtrata primitive	Feasibility
UC8	Agents that publish forever	AI	Agent skill + permanent portfolios	Now — agent-readability gaps
UC9	The arcade that cannot be delisted	Games	Recursive game inscriptions + score attestations	Now
UC10	Composable game assets	Games	Dependency graphs + CORS-open bytes	Near — engine loader
UC11	The digital will	Legacy	Permanent objects + wallet-native transfer	Moonshot — trigger oracle
UC12	The unforkable moment strategy	Attention	Single-artifact rescues	Now

Two patterns in this map shape everything below. First, the decisive primitive is almost never storage: it is the graph (lineage, dependencies, replies) or sponsorship — the two capabilities no competing permanence stack combines with wallet-native ownership. Second, every “now” row is a single-artifact news story waiting to be executed, which is why UC12 exists as a meta-program that sequences the other eleven. The four non-“now” rows each lack exactly one piece — a contract, a pilot, a loader, an oracle — and the launch sequencing in Chapter 5 treats those gaps as build orders, not blockers.

3.1 Music and Culture

Music carries the freshest wound in this report: the two flagship on-chain music platforms are dead or dying, and both died with their media off-chain. It is also where Xtrata’s tooling is most complete — the founder already shipped Audionals, a protocol for producing music directly on Bitcoin⁵⁰.

3.1.1 UC1 “The Album That Outlives the Label”: full albums inscribed with artwork, lyrics, credits as connected objects; sponsored free drops as distribution; pain: catalogs removed from streaming (UMG×TikTok 2024, De La Soul 34-year absence); headline: “This record can never be taken down”; feasibility: now (XTRATA FM + wizard exist)

The pain. Music vanishes from the dominant access layer routinely, and rarely because the artist wants it gone. Universal Music Group — the world’s largest music

⁵⁰ Leather. <https://app.leather.io/support/guide/what-are-audionals>

company — let its TikTok license expire on January 31, 2024, and its entire recorded and publishing catalog went dark on the platform for roughly three months⁵¹. De La Soul’s first six albums spent 34 years off streaming because the samples had never been cleared for digital release⁵². Meanwhile the long tail earns almost nothing even when it stays up: 93.2 million tracks received ten or fewer plays in all of 2024⁵³. The platforms built to fix this then died with their media off-chain. Sound.xyz, which raised \$25 million and paid artists about \$6 million, went offline on January 16, 2026, its farewell note pointing collectors to “decentralized storage” while the listening product itself evaporated^{54 55}. Nina Protocol — the permanence-messaging pioneer — announced its shutdown on May 28, 2026 and goes dark after July 22, 2026^{56 57}. Catalog, the niche survivor, keeps its audio alive only through platform-paid IPFS pinning: its records’ durability depends on the company continuing to exist⁵⁸.

The Xtrata unlock. A full album inscribed as connected objects: each track as chunked immutable media up to 32 MiB, artwork, lyrics, and credits as linked objects, and a self-contained HTML player referencing the audio through same-origin `/i/{id}` recursion⁵⁹. The Suno wizard already builds this player in the browser — audio, cover art, lyrics, one payment⁶⁰ — and a 4.53 MB MP3 already streams from chain data with HTTP range support⁶¹. Distribution runs through sponsored drops: the artist’s deposit covers the network fee, so fans claim a permanent, wallet-owned copy with zero STX⁶². Bitcoin L1 cannot match this: inscriptions above ~100 KB need private miner coordination⁶³, a megabyte costs an estimated \$160–315 even at today’s quiet fees⁶⁴, and Quantum Cats paid \$66,000 for 10 MB⁶⁵. Arweave and IPFS store bytes but no player, ownership, or lineage; the streaming platforms are themselves the deletion risk.

⁵¹ PR Newswire. <https://www.prnewswire.com/news-releases/universal-music-group-agreement-with-tiktok-to-expire-on-january-31-2024-302048634.html>

⁵² Variety. <https://variety.com/2021/music/news/de-la-soul-streaming-services-tommy-boy-1234988719/>

⁵³ Gearnews. <https://www.gearnews.com/spotify-streaming-report-2024-tech/>

⁵⁴ sound.xyz. <https://sound.xyz>

⁵⁵ Billboard. <https://www.billboard.com/pro/sound-xyz-nft-platform-20m-andreessen-horowitz-snoop-dogg/>

⁵⁶ Crypto Briefing. <https://cryptobriefing.com/solana-music-launch-spotify-challenger/>

⁵⁷ Nina Protocol. <https://www.ninaprotocol.com>

⁵⁸ Decentral. <https://www.decentral.io/articles/do-music-nfts-still-matter-my-journey-with-catalog-to-find-out>

⁵⁹ Xtrata (stxtrata/xtrata). <https://github.com/stxtrata/xtrata/blob/main-staging/xtrata-1.0/docs/xtrata-inscription-handbook.md>

⁶⁰ Xtrata. <https://xtrata.xyz/wizard/suno>

⁶¹ Xtrata. <https://xtrata.xyz/i/312>

⁶² Hiro Mainnet API.

<https://api.mainnet.hiro.so/v2/contracts/interface/SP3JNSEXAZP4BDSHV0DN3M8R3P0MY0EEBQQZX743X/xtrata-drops-v1-0>

⁶³ b10c.me. <https://b10c.me/observations/09-non-standard-transactions/>

⁶⁴ mempool.space. <https://mempool.space/api/v1/fees/recommended>

⁶⁵ The Defiant. <https://thedefiant.io/news/nfts-and-web3/quantum-cats-ordinals-collection-pays-homage-to-satoshi>

The headline. *“This record can never be taken down.”*

Feasibility. Now. XTRATA FM, the drops contract, and the inscription wizard all exist; the only remaining work is artist recruitment and the drop itself — ideally timed to the next high-profile streaming takedown.

3.1.2 UC2 The Contract-Enforced Remix Lineage Economy: parent escrow at mint makes derivation machine-enforceable; pain: sample clearance \$500–50k and 2–6 months, MLC unmatched-royalty black box ~\$397M–\$1B; headline: “The remix economy where every sample pays its ancestor automatically”; feasibility: near (needs royalty-split contract)

The pain. Derivation is the music industry’s most expensive metadata failure. Clearing a single sample costs \$500–\$50,000+ and typically takes two to six months⁶⁶; clearing De La Soul’s catalog for streaming took a specialist team a full year, with some samples replayed because clearance was uneconomic⁶⁷. When attribution fails, money strands: the US Mechanical Licensing Collective (MLC) sits on an estimated ~\$397 million in unmatched royalties — outside estimates approach \$1 billion — because the metadata is wrong, and unmatched funds default to distribution by market share, i.e., to the biggest publishers^{68 69}. Disputes scale to career size: the “Blurred Lines” case closed at \$4,983,766.85 plus 50% of future royalties⁷⁰, and even web3-native releases repeat the failure — 3LAU’s \$11.7 million Ultraviolet auction drew a lawsuit from his co-writer, who said she was offered a one-time \$25,000⁷¹. Today attribution lives in liner notes, PRO filings, and a fan wiki that De La Soul’s own clearance expert calls inaccurate⁶⁷.

The Xtrata unlock. Parent escrow at mint: to inscribe a remix as a child object, the minter must own the parent, and the contract escrows that parent during the mint — multi-parent supported, with the resulting lineage stored as chain state other contracts can read and act on⁷². Wire revenue splits through the market layer — the live collection-mint contract already splits mint proceeds between artist, marketplace, and operator⁷³ — and every remix pays its ancestors automatically. Bitcoin Ordinals

⁶⁶ Chartlex. <https://www.chartlex.com/blog/business/sample-clearance-guide-musicians-2026>

⁶⁷ Okayplayer. <https://www.okayplayer.com/heres-how-de-la-soul-cleared-the-samples-for-their-classic-catalogs-streaming-debut/369919>

⁶⁸ Interspace Music. <https://interspacemusic.com/blog/the-mlc-has-paid-out-billion-the-money-stuck-in-its-black-box-gets-split-by-market-share/>

⁶⁹ Complete Music Update. <https://completemusicupdate.com/the-mlcs-billion-dollar-black-box-in-the-spotlight-in-us-copyright-office-review/>

⁷⁰ Entertainment Weekly. <https://ew.com/music/2018/12/13/blurred-lines-copyright-lawsuit-robin-thicke-pharrell-williams-pay/>

⁷¹ Billboard. <https://www.billboard.com/pro/3lau-nft-ultraviolet-auction-songwriter-sues-share-profits/>

⁷² Xtrata (stxtrata/xtrata). <https://github.com/stxtrata/xtrata/blob/main-staging/xtrata-1.0/docs/recursive-inscriptions.md>

⁷³ Xtrata (stxtrata/xtrata). <https://github.com/stxtrata/xtrata/blob/main-staging/xtrata-1.0/docs/contract-inventory.md>

does have parent/child provenance, but it lives in witness tags only off-chain indexers read; Bitcoin script cannot act on it, and Ordinals has no typed dependency or reply relations at all^{74 75}. The proving demo: a 100-remix on-chain relay where every mint is a visible, payable node in the family tree.

The headline. *“The remix economy where every sample pays its ancestor automatically.”*

Feasibility. Near. The lineage primitives are live today; the gap is one royalty-split contract wired to the market escrow — plus the honest caveat that on-chain lineage records provenance, not legal clearance, and should be positioned as attribution rails, not licensing replacement.

3.1.3 UC3 The Radio Station That Cannot Be Shut Down: XTRATA FM as a public, SDK-embeddable permanent broadcaster; any site can embed the same station; headline: “The first radio station with no server to raid”; feasibility: now

The pain. Every music broadcaster dies with its operator. Catalog ran “Catalog Radio,” a live 24/7 broadcast programmed from its on-chain catalog — it disappeared into the platform’s niche economics⁵⁸. Sound’s player is gone; Nina’s is going^{54 56}. Conventional broadcast adds its own chokepoints: playlist servers, app-store gatekeepers, licensing boards. A permanence medium should route around all of these, and none of the dead or dying platforms ever did, because their media and players were never on-chain.

The Xtrata unlock. XTRATA FM is already live: a widget that reads the chain’s playable index and streams multi-megabyte MP3s directly from inscription bytes over HTTP range requests — in the code’s own words, “THIS RADIO HAS NO PLAYLIST SERVER”^{76 77}. The byte endpoints are CORS-open and cache-immutable, so any website can embed the same station via the SDK. And because objects reconstruct independently of xtrata.xyz under a published public proof standard, taking down the company would silence nothing⁷⁸. Catalog Radio proved the 24/7 format but kept media on platform-paid IPFS; XTRATA FM’s media is the chain itself. Add block-height-triggered programming — every Bitcoin block picks the next song — and the station becomes a chain-native artifact, not a web app.

The headline. *“The first radio station with no server to raid.”*

⁷⁴ Ordinal Theory Handbook. <https://docs.ordinals.com/inscriptions/provenance.html>

⁷⁵ Ordinal Theory Handbook. <https://docs.ordinals.com/inscriptions/recursion.html>

⁷⁶ Xtrata. <https://xtrata.xyz/assets/main-BuSmoc1z.js>

⁷⁷ Xtrata. [https://xtrata.xyz/index/playable?](https://xtrata.xyz/index/playable?contract=SP3JNSEXAZP4BDSHV0DN3M8R3P0MY0EEBQQZX743X.xtrata-v3-2-3)

[contract=SP3JNSEXAZP4BDSHV0DN3M8R3P0MY0EEBQQZX743X.xtrata-v3-2-3](https://xtrata.xyz/index/playable?contract=SP3JNSEXAZP4BDSHV0DN3M8R3P0MY0EEBQQZX743X.xtrata-v3-2-3)

⁷⁸ Xtrata (stxtrata/xtrata). <https://github.com/stxtrata/xtrata/blob/main-staging/xtrata-1.0/docs/reconstruction-spec.md>

Feasibility. Now. The gap is distribution, not code: XTRATA FM currently has zero independent coverage, so the stunt is the embed-anywhere push — a hundred indie sites streaming the identical unstoppable station on the same day.

3.2 Truth and Evidence

Permanence matters most where deletion is deliberate. These three use cases convert Xtrata from a media platform into public infrastructure — and sponsored transactions (a third party prepays the network fee) are what make them usable by people who have never touched crypto.

3.2.1 UC4 The War-Crime Evidence Vault: sponsored claims let NGOs fund permanence for witnesses who have no crypto; hash-verified bundles meet Berkeley Protocol needs; headline: “The evidence YouTube deleted is now undeletable”; feasibility: now (drops + inscriptions), partnerships needed

The pain. Platforms delete atrocity evidence at industrial scale. YouTube’s automated removals erased an estimated 120,000–150,000 Syrian-conflict videos — material the UN-published WITNESS submission calls “some of the best documentation of war crimes and human rights abuses in Syria” — with prominent channels terminated up to five times⁷⁹. The Berkeley Protocol, the global evidence standard used by Ukrainian prosecutors, demands exactly what the upload moment lacks: capture before deletion, cryptographic integrity hashing, chain of custody⁸⁰. Legal compulsion deletes journalism too: in September 2025 an Indian court granted an ex parte order forcing named journalists to delete articles within 36 hours and letting Adani Enterprises flag further URLs for takedown without court vetting⁸¹. And the fallback archive is fragile — the Internet Archive itself was breached and knocked into read-only mode in October 2024⁸².

The Xtrata unlock. An evidence bundle as an object: raw footage as chunked media whose incremental SHA-256 chain hash is computed at inscription time — integrity is protocol-native, not bolted on — with geolocation, verification notes, and translations as dependency-linked children and prosecutor annotations as permanent reply threads^{59 49 78}. The decisive feature is sponsored claims: an NGO funds a drop, and a witness in a conflict zone with no crypto, no bank account, and no safe identity deposits evidence permanently, for free⁶². Arweave offers no sponsored claims, no

⁷⁹ WITNESS. <https://www.ohchr.org/Documents/Issues/Opinion/ContentRegulation/Witness.pdf>

⁸⁰ UC Berkeley Human Rights Center. <https://humanrights.berkeley.edu/berkeley-protocol-digital-open-source-investigations>

⁸¹ Inventiva. <https://www.inventiva.co.in/stories/the-game-of-defamation-in-the-biggest-democratic-nation-when-telling-the-truth-gets-you-a-lawsuit/>

⁸² The Verge. <https://www.theverge.com/2024/10/14/24269741/internet-archive-online-read-only-data-breach-outage>

ownership, and no threads; YouTube is the deletion mechanism; NGO servers are funding-fragile single points that strip native provenance when they re-host.

The headline. *“The evidence YouTube deleted is now undeletable.”*

Feasibility. Now — drops and inscriptions are live. The gap is institutional: one NGO partnership (Syrian Archive, Mnemonic, or WITNESS) and a published abuse-policy position, because unmoderatable permanence cuts both ways and must be answered before launch, not after.

3.2.2 UC5 The Civic Data Rescue: archived copies of deleted public datasets as queryable on-chain objects; headline: “The dataset the government deleted that can never be deleted again”; feasibility: now, one-object news story

The pain. In early 2025 the US government demonstrated that public data is politically perishable: more than 8,000 pages across a dozen-plus federal sites were taken down in days⁸³; over 2,000 datasets disappeared from data.gov; globalchange.gov shut down and climate.gov stopped publishing after its staff was terminated^{84 85}. Courts eventually vacated the removal directives — but restorations arrived partial and captioned with political disclaimers⁸⁶. This sits atop background decay: 25% of all webpages that existed between 2013 and 2023 are already gone, and 21% of government pages contain at least one broken link⁸⁷.

The Xtrata unlock. A rescued dataset inscribed as a queryable object — versions as children, third-party mirrors referencing the canonical copy through existence-only dependencies, and alteration disputes (“this version was edited”) attached as reply threads that can never be detached from the object. Sponsored claims turn rescueathons — the volunteer datathons that scrambled to copy federal data in January 2025 — into zero-cost mass inscription events⁶². The Wayback Machine honors removals and can be knocked offline; court-ordered restoration depends on who holds power; IPFS pins survive only while someone keeps paying. An inscribed object answers to none of these, and its Bitcoin-anchored timestamp doubles as proof of what the data said at time T.

The headline. *“The dataset the government deleted that can never be deleted again.”*

⁸³ Society of Environmental Journalists (citing NYT analysis).

<https://www.sej.org/headlines/thousands-us-government-web-pages-have-been-taken-down-friday>

⁸⁴ Freie Universität Berlin (Earth Sciences Library).

<https://www.geo.fu-berlin.de/en/bibliotheken/Aktuelles/Verschwindende-Daten---Trump-Administration-entfernt-Klimainformationen-von-Regierungswebsites.html>

⁸⁵ NPR. <https://www.npr.org/2025/08/08/nx-s1-5495338/climate-change-environment-websites-trump>

⁸⁶ Fierce Healthcare. <https://www.fiercehealthcare.com/regulatory/judge-vacates-trump-administrations-removal-health-web-pages>

⁸⁷ ISSN (Pew Research Center study summary). https://www.issn.org/newsletter_issn/when-online-content-disappears/

Feasibility. Now — a single well-chosen object (one deleted CDC surveillance dataset, or the data behind the removed CEJST environmental-justice tool) is a self-contained news story. The only scoping constraint is the 32 MiB object cap, which simply means targeting high-value, small-footprint datasets first.

3.2.3 UC6 The Self-Correcting Scientific Record: papers/data inscribed; retraction notices as on-chain replies that can never be detached; pain: 82% of retracted papers keep being cited, data availability decays 17%/year; headline: “The paper that carries its own retraction”; feasibility: near

The pain. The scientific record fails in both directions. Underlying data decays at roughly 17% per year — broken email addresses and obsolete storage are the main killers⁸⁸ — and even under modern mandates, only about half of papers have actually accessible data⁸⁹. Meanwhile corrections detach from the work: 82% of retracted biomedical papers continue to be cited, and only 4–6% of those citations acknowledge the retraction⁹⁰. The record neither survives nor updates itself.

The Xtrata unlock. Paper, dataset, code, and replication as linked objects — authorship proven through parent/child ownership links, citations and replications recorded as existence-only dependencies — with the retraction or erratum inscribed as an on-chain reply to the original object. A reply cannot be detached, silently edited, or lost in a metadata-propagation failure: any tool that touches the paper touches its retraction. Journals, funders, or universities sponsor inscription so authors in the global south pay nothing⁶². Repositories approximate this with DOI glue and inconsistent retraction metadata; here the correction is a property of the object itself. Timestamped preregistrations — hypotheses inscribed before results exist — come free with the same primitive.

The headline. *“The paper that carries its own retraction.”*

Feasibility. Near. The primitives are live; the gap is one journal or funder pilot and a reader-side interface that surfaces reply threads prominently enough to change citing behavior.

3.3 AI, Agents, and Provenance

Synthetic media has made provenance the internet’s central question. Xtrata answers it in both directions — proving the human and anchoring the machine.

⁸⁸ Tampere University. https://research.tuni.fi/uploads/2020/10/1e12eb53-research-data-management_basics_20200104_v2.pdf

⁸⁹ bioRxiv. <https://www.biorxiv.org/content/10.1101/2025.02.06.636914v3.full>

⁹⁰ Journal of Modern Medical Information Sciences. <https://journals.sbm.ac.ir/jmlis/article/download/48705/35366/254299>

3.3.1 UC7 The Certified-Human Content Registry: wallet-signed, immutable publication as the anchor C2PA lacks; “made by a human, provably, forever”; headline: “The last place on the internet where human-made means something”; feasibility: now

The pain. Deepfake-enabled fraud caused a reported \$1.5 billion-plus in losses worldwide in the first nine months of 2025 alone⁹¹, and unaided humans identify high-quality deepfakes only about 24.5% of the time — below chance⁹². The standards answer, C2PA Content Credentials (cryptographically signed provenance metadata embedded in the file), now ships in major cameras, phones, and creative tools — but social platforms strip embedded metadata, C2PA manifests included, during upload and transcoding. The proof dies exactly where the fakes spread⁹³. Regulators have noticed: the EU AI Act’s machine-detectable marking requirement for AI output applies from August 2, 2026, and the official guidance prescribes watermarking and logging *alongside* metadata precisely because metadata is easily removable⁹⁴. Detection is losing the arms race; positive provenance needs an anchor that cannot be stripped.

The Xtrata unlock. A creator signs and inscribes the work itself — content, hash, timestamp, wallet identity resolvable to a .btc name — as an immutable object. No re-encode, screenshot, or platform upload can detach the registry entry, because it lives in chain state rather than in file metadata. Content-addressed dedupe makes “has this exact work been registered before?” a single read call, and the native market lets provenance-rich human work command a premium. This is a complement to C2PA, not a competitor: inscribe the manifest’s hash and the Content Credential gains the tamper-proof registry of record it currently lacks. Hash-only timestamping services prove a hash existed but leave the media on perishable infrastructure; Xtrata anchors content and proof together.

The headline. *“The last place on the internet where human-made means something.”*

Feasibility. Now. Minting, wallet signing, and dedupe are live; the product work is a one-click “certify human-made” flow in the wizard and a verification badge any site can render.

⁹¹ Veriff. <https://www.veriff.com/fraud/deepfake-fraud-cost-2026>

⁹² Adaptive Security. <https://www.adaptivesecurity.com/blog/ai-deepfake-trends-the-complete-2025-2026-guide-to-statistics-threats-detection-and-defense-strategies>

⁹³ AIBuzz. <https://aibuzz.blog/ai-watermarking-vs-metadata-vs-fingerprinting/>

⁹⁴ Truescreen. <https://truescreen.io/articles/c2pa-standard-history-limitations/>

3.3.2 UC8 Agents That Publish Forever: aibtc agents minting their outputs, portfolios, and memory as Xtrata objects; agent-to-agent commerce in sBTC/USDCx; headline: “The first AI agents with permanent bodies of work”; feasibility: now (agent skill ships) — but needs llms.txt, MCP server, drops/market coverage in skill docs

The pain. The agent economy has identity and payments but no body of work. On Stacks, aibtc agents already hold self-custodial wallets, message each other for 100 sats of sBTC, and register ERC-8004 identities — an on-chain agent identity and reputation standard — with registries deployed on mainnet⁹⁵ ⁹⁶. The x402 payment protocol has processed roughly 165 million agent transactions, though much of that is still testing rather than genuine commerce⁹⁷. Yet everything an agent produces — reports, art, code, analysis, trading records — evaporates with the session or the host. An agent with an ERC-8004 identity today holds a résumé that says nothing and cannot be verified.

The Xtrata unlock. Xtrata ships a 1,106-line agent skill that explicitly teaches aibtc agents to mint autonomously: chunk the payload, compute the chain hash, dedupe against the canonical registry, seal, verify — with deny-mode post-conditions and deterministic spend caps as safety rails⁹⁸. Every output becomes a permanent object bound to the agent’s wallet: a portfolio no platform can delete, salable to other agents in sBTC or USDCx through the escrow markets, and claimable through sponsored drops so a newborn agent needs no STX to start. Botto proved machine authorship sells — over \$5 million in sales and a Sotheby’s solo show⁹⁹ — but Botto’s works live behind conventional NFT pointers; an aibtc agent’s corpus would live in Bitcoin-anchored chain state, traversable as a graph by the reputation and evaluator contracts the ecosystem is already deploying.

The headline. “*The first AI agents with permanent bodies of work.*”

Feasibility. Now — the skill doc ships and the autonomous loop is documented end to end. The gaps are agent-readability polish: a real llms.txt (today the route serves the app shell), an Xtrata MCP server of its own, and skill-doc coverage of the drops, market, and reply-thread functions that exist on-site but not yet in the agent docs.

⁹⁵ aibtc. <https://aibtc.com/llms.txt>

⁹⁶ aibtcdev. <https://github.com/aibtcdev/erc-8004-stacks>

⁹⁷ RZLT. <https://www.rzlt.io/blog/agentic-payments-2026-x402-explainer>

⁹⁸ Xtrata (stxtrata/xtrata).

https://github.com/stxtrata/xtrata/blob/OPTIMISATIONS/xtrata-1.0/XTRATA_AGENT_SKILL.md

⁹⁹ Botto. <https://botto.com/autonomous-ai-artist-botto-350k-sales-sothebys>

3.4 Games and Play

Gamers are the most mobilized permanence constituency alive — the only one that has already forced a legislature to pay attention.

3.4.1 UC9 The Arcade That Cannot Be Delisted: whole games as recursive inscriptions; arcade-scores contract with secp256k1 attestations = permanent global leaderboards; pain: The Crew erasure, 87% of classic games unavailable, 1.3M-signature Stop Killing Games; headline: “The game that will still boot in 100 years”; feasibility: now

The pain. Ubisoft did not merely delist The Crew: it killed the servers in March 2024, then revoked the license from paying customers’ libraries without refunds — triggering a US class action, a lawsuit from France’s leading consumer group, and the Stop Killing Games citizens’ initiative, which passed 1.3 million verified signatures and forced a mandatory European Commission review^{100 101}. The baseline loss rate is worse than silent film: 87% of pre-2010 US games are commercially unavailable¹⁰². Konami made P.T. — a free demo — impossible to reacquire, and PS4 consoles with it installed sold for \$1,000–1,500¹⁰³. Flash’s end-of-life orphaned an entire creative era; 200,000-plus games and animations now survive only through volunteer archivists operating in a copyright gray zone¹⁰⁴.

The Xtrata unlock. A whole game as a recursive inscription: HTML, JavaScript, and assets chunked on-chain, booting straight from `/i/{id}` — a 697 KB on-chain application already serves exactly this way¹⁰⁵. The arcade-scores contract adds what no delisted game can have: permanent global leaderboards with secp256k1-signed score attestations (secp256k1 is Bitcoin’s own signature scheme), nonces, and replay protection⁷³. Ownership is wallet-native, so no publisher can revoke a token from a player’s library. On Bitcoin L1 this pattern collapses at scale — one on-chain Ordinals game needed roughly two million indexer round-trips to compute state for a 10,000-item collection¹⁰⁶ — because Bitcoin script cannot read inscription relationships. Xtrata’s graph is legible to contracts directly. The honest scoping: inscribe your own IP or partner with rights-holders; Xtrata cannot legalize hosting someone else’s game.

The headline. “*The game that will still boot in 100 years.*”

¹⁰⁰ GameDeveloper (Reuters). <https://www.gamedeveloper.com/business/french-consumer-group-sues-ubisoft-over-shutdown-of-the-crew>

¹⁰¹ Eurogamer. <https://www.eurogamer.net/ubisoft-sued-over-controversial-the-crew-shutdown>

¹⁰² Video Game History Foundation. <https://gamehistory.org/87percent/>

¹⁰³ GamingBolt. <https://gamingbolt.com/remembering-p-t-the-incredible-horror-teaser>

¹⁰⁴ Boing Boing. <https://boingboing.net/2025/06/16/old-adobe-flash-games-preserved-by-flashpoint-archive.html>

¹⁰⁵ Xtrata. <https://xtrata.xyz/i/1107>

¹⁰⁶ ordinals/ord (GitHub issue #3719). <https://github.com/ordinals/ord/issues/3719>

Feasibility. Now. The contracts and a working on-chain app exist; the gap is one indie studio shipping its game as an inscription with a live leaderboard.

3.4.2 UC10 Composable Game Assets with Living Provenance: assets as objects with dependency graphs, loadable into games that didn't exist at mint time; headline: "The sword that outlives its game"; feasibility: near

The pain. Game assets are born captive: when the game dies, the sword dies with it — The Crew's players lost paid-for content the moment the servers closed¹⁰⁰. Even while games live, items cannot leave their walled gardens, and provenance — who made the asset, which game it came from, which build is authentic — dies with the publisher's servers. Torrents and fan archives preserve bytes but no lineage and no ownership.

The Xtrata unlock. An asset as an object with a declared dependency graph: mesh, texture, and audio chunks as dependencies, the artist recorded as creator, ownership as a SIP-009 token (Stacks' NFT standard) tradable in sBTC or USDCx. Because dependencies are existence-only and permissionless — anything can reference anything — a game that did not exist when the asset was minted can still load it over the CORS-open `/i/{id}` byte endpoints and verify its lineage on-chain^{72 59}. Conventional NFTs point at off-chain files that rot; Ordinals assets cannot be read by game-logic contracts at all. Here the asset's provenance stays alive and machine-readable as it moves between engines, mods, and remasters — with the original artist visible at every hop, and optional market royalties wired through the same escrow contracts that settle everything else on the platform⁴⁹.

The headline. *"The sword that outlives its game."*

Feasibility. Near. Objects, dependencies, and markets are live; the gap is one engine-side loader (an SDK bridge for web engines, Unity, or Godot) and a partner game willing to prove cross-title assets in public.

3.5 Moonshots

Two entries close the chapter: one that requires a layer Xtrata does not yet have, and one that requires nothing but nerve.

3.5.1 UC11 The Digital Will: encrypted legacy objects with dead-man's-switch triggers; pain: iCloud court orders, licenses dying at death, ~4M lost BTC; headline: "The inheritance that executes itself"; feasibility: moonshot (oracle/trigger layer)

The pain. Digital death is bureaucratic chaos. Without a pre-configured legacy contact, families need a court order to reach a deceased person's iCloud photos and

messages — “a process that can take months and is not always successful”¹⁰⁷. Purchased media dies with the buyer: as Australia’s eSafety Commissioner puts it, “you may have just bought a licence for the term of your life”¹⁰⁸. An estimated four million bitcoins are lost forever, much of that through death and lost keys¹⁰⁹. Dead-man’s-switch services (tools that auto-release files if you stop checking in) exist, but they concentrate your most sensitive documents in one startup that must outlive you — and attackers already exploit the inheritance flow itself, with phishing campaigns faking death notices against password-manager legacy requests¹¹⁰.

The Xtrata unlock. Encrypted legacy objects — letters, photo archives, key shards, instructions — inscribed permanently and organized as an estate tree: person to documents to assets to instructions, with wallet-native transfer to heir wallets or multisig arrangements instead of petitions to Apple with a death certificate. Sponsored claims let an estate lawyer set up a client’s vault without the client ever touching crypto. What Xtrata cannot do is the trigger: there is no on-chain dead-man’s switch. The honest architecture keeps permanence and ownership on-chain and puts the trigger — check-ins, death-certificate attestation — in an oracle/service layer that releases decryption keys. That separation is a feature: the trigger service can fail or be replaced without ever endangering the underlying objects.

The headline. *“The inheritance that executes itself.”*

Feasibility. Moonshot. Storage, graph, and ownership primitives exist today; the missing layer is the trigger oracle plus legal recognition, which makes this a partner product — estate-services firms as sponsors — rather than a core-protocol feature.

3.5.2 UC12 The Unforkable Moment Strategy: a sequence of single-artifact publicity stunts — rescue one famous deleted thing at a time; each object is a self-contained news story; feasibility: now, marketing program

The pain this solves is attention. Abstract permanence does not trend; rescued artifacts do. The record proves it: French Montana inscribing a single song across an entire Bitcoin block made mainstream press¹¹¹, and a Super Nintendo emulator inscribed on Bitcoin rode the preservation story — 87% of classic games at risk of disappearing — into coverage far beyond crypto media^{102 112}. Neither stunt delivered

¹⁰⁷ Legacy Options. <https://www.legacyoptions.com/post/managing-digital-accounts-after-a-death>

¹⁰⁸ eSafety Commissioner (Australia). <https://www.esafety.gov.au/key-topics/digital-wellbeing/what-happens-to-your-digital-accounts-after-you-die>

¹⁰⁹ Fortune (China), citing Chainalysis.

http://www.fortunechina.com/investing/c/2018-10/22/content_317942.htm

¹¹⁰ Malwarebytes. <https://www.malwarebytes.com/blog/news/2025/10/phishing-scam-uses-fake-death-notices-to-trick-lastpass-users>

¹¹¹ Bitcoinist. <https://bitcoinist.com/bitcoin-french-montana-unreleased-song-ordinals/>

¹¹² ForkLog. <https://forklog.com/en/ninjalerts-team-launches-super-nintendo-emulator-on-bitcoin-blockchain/>

actual permanence with ownership; both still earned the headline. Xtrata can run the same play with the substance included.

The program. A sequenced calendar of single-artifact rescues, each chosen because it is famous, deleted or suppressed, emotionally resonant, and rights-clear — one object, one headline, one news cycle, on a monthly cadence. An illustrative first arc, matching the Chapter 5 sequence: Month one, the undeletable album with a heritage act whose catalog was label-orphaned — the De La Soul scenario, this time with permanence instead of a 34-year wait — dropped straight into the Nina shutdown discourse⁵². Month two, a deleted CDC surveillance dataset rescued as a queryable object — federal data carries no rights friction, and the 2025 purges are still fresh⁸³. Month three, a war-crime evidence bundle with an NGO partner⁷⁹. Further out, an indie studio’s delisted game re-released as a bootable inscription with a live global leaderboard. Each stunt is engineered to be self-contained press: the artifact is the story, the inscription transaction is the dateline, and the object remains online as permanent proof that the claim in the headline is true.

The headline. Each object writes its own — *“The dataset the government deleted that can never be deleted again”* — with the program itself earning the meta-headline: *“Every month, one famous deleted thing comes back forever.”*

Feasibility. Now — this is a marketing program, not an engineering project. The gap is a rights-clearance pipeline and roughly one partnership per stunt: the first month needs a single artist from the founder’s own music network, and the second — federal data — needs no partner at all.

Twelve use cases, eight of them buildable on contracts already live — but “buildable” is not “prioritized,” and every “now” carries real constraints that this chapter has deliberately left attached to each case: per-megabyte cost against Arweave, the 32 MiB object cap, zero independent coverage of XTRATA FM, the abuse-policy question that unmoderatable permanence raises, and the legal caveats on lineage and IP. Chapter 4 prices those constraints honestly; Chapter 5 converts this map into a launch sequence.

4. Honest Constraints: What Could Undermine All of This

The use cases in Chapter 3 survive contact with chain data. What could still sink them is structural, not technical. Three constraints — cost positioning, platform dependence, missing attention — each carry a mitigation that directly shapes the launch sequence in Chapter 5.

4.1 Cost and Scale Realism

4.1.1 Arweave 5–100× cheaper per MB — Xtrata must own high-value relationship-rich objects, not bulk storage

Xtrata loses any contest scored in dollars per megabyte. Arweave stores a megabyte permanently for roughly \$0.019 one-time¹¹³; Xtrata’s chunked inscription pipeline works out to an estimated \$0.10–2.00 per MB at current Stacks fee levels¹¹⁴ — a 5–100× handicap, and a flattering one, since Arweave’s figure is measured while Xtrata’s is extrapolated from live fee samples. Arweave’s 200-year permanence is itself an economic projection rather than a guarantee¹¹⁵, but it is a cheap projection. The implication is strategic, not fatal: Xtrata must never compete as bulk storage. Its win condition is objects whose value lies in ownership, typed relationships, and Bitcoin-anchored finality — music masters, evidence bundles, provenance records — because Arweave stores bytes with no native ownership, market, or object graph attached¹¹⁶. Chapter 3’s use cases were selected so that question never comes first.

4.2 Platform Dependencies

4.2.1 Stacks 2026 chain-state-pruning proposal as long-horizon reconstruction risk; single-founder bus factor; SDK not on npm; missing llms.txt/MCP server

Xtrata inherits Stacks’ roadmap, and one live proposal brushes against its founding promise. A 2026 core-developer proposal would cap at-block lookback at six cycles and let nodes prune older chain state, leaving full history to archive nodes¹¹⁷. If adopted, very-long-horizon reconstruction of Xtrata objects could come to depend on archive operators — a milder echo of the indexer dependence Xtrata criticizes in rivals. The proposal is still at discussion stage and its impact on Xtrata’s storage layout is unverified, but “independently rebuildable” claims should be stress-tested against it before they headline a campaign.

The second dependency is human. Engineering runs through a single GitHub account with 1,557 contributions in the past year¹¹⁸; the only other named human contributor shows zero¹¹⁹. One founder plus AI assistants is a bus factor of one. The third dependency is an unfinished builder surface: the SDK ships only as unpublished

¹¹³ kkdemian (citing ViewBlock). <https://kkdemian.com/blog/arweave-ar-permanent-storage-endowment-demand-risk>

¹¹⁴ Hiro. <https://api.mainnet.hiro.so/extended/v1/tx>

¹¹⁵ ArDrive (Arweave ecosystem). <https://ardrive.io/what-is-arweave>

¹¹⁶ ChainScore Labs. <https://chainscorelabs.com/comparisons/storage-ipfs-vs-arweave-vs-filecoin/storage-token-economics-models/filecoins-deal-making-vs-arweaves-endowment-model>

¹¹⁷ Stacks Forum. <https://forum.stacks.org/t/chain-state-pruning-and-at-block-proposed-change/18685>

¹¹⁸ GitHub. <https://github.com/stxtrata>

¹¹⁹ GitHub. <https://github.com/shubh2294>

v0.1.0 workspace packages, not on npm¹²⁰; `/llms.txt` returns the app shell instead of agent-readable content¹²¹; and Xtrata has no MCP server of its own in an ecosystem where aibtc.com already ships `llms.txt`, skill files, and MCP tooling as table stakes¹²². All three gaps are cheap to close — weeks, not quarters — which is precisely why they gate everything else.

4.3 Traction Risk

4.3.1 Zero press, tiny community — why the stunt strategy (UC12) is the bridge from tech to attention

The gravest constraint is attention. Xtrata has 117 X followers¹²³, no Discord or Telegram presence¹²¹, zero press beyond an automated aixbt.tech listing and Gamma indexing¹²⁴, and a market showing no live listings after seven lifetime asks¹²⁵. Even the founder's marquee credential — Audionals' TRUTH selling out in about an hour — is self-reported and independently unverified¹²⁶. A platform whose demand is mostly team mints cannot negotiate partnerships from strength. That is exactly why UC12 leads the go-to-market: one rescued, culturally famous artifact is a self-contained news story that manufactures the attention the project cannot yet buy, and every sponsored claim converts spectators into users.

These constraints write the Chapter 5 sequence: fix the developer-agent surface first because it is cheap, fire the stunts second because attention unlocks everything else, and schedule the heavy builds — remix royalties, institutional evidence vaults — for when an audience exists to notice them.

5. The Launch Sequence: What To Do First

The whole report collapses into one sequence: fix the surface that machines and developers touch, then manufacture attention, then build the economy that monetizes it. The order is not optional — the stunts fail if the surface is broken, and the economy fails if nobody is watching. With 117 followers and zero press, attention cannot be bought, so it must be manufactured with artifacts^{127 128}. And one deadline is

¹²⁰ GitHub — stxtrata/xtrata. <https://github.com/stxtrata/xtrata/blob/main-staging/xtrata-1.0/docs/documentation-index.md>

¹²¹ Xtrata. <https://xtrata.xyz>

¹²² AIBTC. <https://aibtc.com/llms.txt>

¹²³ X. <https://x.com/XtrataLayers>

¹²⁴ aixbt.tech. <https://aixbt.tech/projects/Xtrata-69cbe7e162fd35aabfaf2e0a>

¹²⁵ Xtrata. <https://xtrata.xyz/market>

¹²⁶ Leather. <https://app.leather.io/support/guide/what-are-audionals>

¹²⁷ X. <https://x.com/XtrataLayers>

¹²⁸ aixbt.tech. <https://aixbt.tech/projects/Xtrata-69cbe7e162fd35aabfaf2e0a>

immovable: Nina Protocol goes offline on July 22, 2026 — the day after this report — making the music stunt a this-week decision, not a this-quarter one¹²⁹.

5.1 Fix the Developer-Agent Surface (Week 0–4)

5.1.1 Publish SDK to npm, ship llms.txt and an Xtrata MCP server, extend agent skill to drops/market/replies — cheap, high-leverage

Four fixes, each measured in days, and all four gate the value of every stunt that follows. First, publish @xtrata/sdk and @xtrata/reconstruction to npm: both exist only as v0.1.0 workspace packages, so a developer who reads a stunt headline cannot install the tool that made it¹³⁰. Second, ship a real llms.txt — the route currently serves the app shell — and an Xtrata MCP server; aibtc.com already treats llms.txt, skill files, and MCP tooling as table stakes, and MCP is the de-facto agent tool layer (~97 million monthly SDK downloads)^{131 132 133}. Third, extend the 1,106-line agent skill beyond mint/transfer/query to cover the drops, market, and reply-thread functions that exist on-site but are absent from the agent documentation¹³⁴. Fourth, finish the Zero Authority DAO bounty — 200 STX across four winners — which is live but pre-launch: deadline TBA, gallery empty, the official-bounty button still showing the developer placeholder “Add the official Zero Authority bounty URL in CONFIG before launch”¹³⁵.

This precedes publicity because every stunt in 5.2 ends in a call to action — claim a copy, verify the hash, build the next object. If the arriving agent or developer cannot act within minutes, attention converts to nothing — and with engineering running through a single founder account, each fix is scoped to what one person can ship in four weeks¹³⁶.

¹²⁹ Nina Protocol. <https://www.ninaprotocol.com>

¹³⁰ GitHub — stxtrata/xtrata. <https://github.com/stxtrata/xtrata/blob/main-staging/xtrata-1.0/docs/documentation-index.md>

¹³¹ Xtrata. <https://xtrata.xyz>

¹³² AIBTC. <https://aibtc.com/llms.txt>

¹³³ Pickaxe. <https://pickaxe.co/post/mcp-vs-a2a-protocol>

¹³⁴ Xtrata (GitHub, stxtrata/xtrata, OPTIMISATIONS branch).

https://github.com/stxtrata/xtrata/blob/OPTIMISATIONS/xtrata-1.0/XTRATA_AGENT_SKILL.md

¹³⁵ Xtrata. <https://xtrata.xyz/masterpiece>

¹³⁶ GitHub. <https://github.com/stxtrata>

5.2 Fire the First Three Stunts (Month 1–3)

5.2.1 Sequence: music drop timed to Nina shutdown discourse → civic data rescue → evidence vault pilot with an NGO partner

Stunt one, this week: the undeletable music drop. Sound.xyz has been offline since January 16, 2026¹³⁷; Nina — roughly 40,000 monthly users at its peak — winds down tomorrow^{129 138}. Recruit one artist from the founder’s music network, inscribe the album as connected objects (tracks, artwork, self-contained player), and distribute through sponsored drops so displaced listeners claim a permanent copy with zero STX. The headline writes itself — *“This record can never be taken down”* — and the discourse window is days wide.

Stunt two, month two: the civic data rescue. One deleted federal dataset — a CDC surveillance file or the data behind a removed climate tool — inscribed as a queryable, versioned object. More than 8,000 pages were purged from US government sites in early 2025 and over 2,000 datasets vanished from data.gov^{139 140}; federal data carries no rights friction, so this stunt needs no partner at all.

Stunt three, month three: the evidence-vault pilot with one NGO partner — Syrian Archive, Mnemonic, or WITNESS. YouTube’s classifiers erased an estimated 120,000–150,000 Syrian war-crime videos¹⁴¹; sponsored claims let a witness deposit evidence permanently, for free. Publish the abuse-policy position before launch, not after — unmoderatable permanence cuts both ways. And launch the bounty properly as its own stunt: filled-in config, a real deadline, a public gallery. In the Stacks scene, a functioning 200 STX creative bounty is itself news¹³⁵.

5.3 Build the Remix Economy (Month 3–9)

5.3.1 Royalty-split contracts on parent/child mints; position XTRATA FM as the venue

The one genuinely new contract on the roadmap is the royalty split: wire revenue shares through the market escrow to parent/child mints so every remix pays its ancestors automatically. The lineage primitive is live — parents are escrowed at mint

¹³⁷ Sound.xyz. <https://sound.xyz>

¹³⁸ Crypto Briefing. <https://cryptobriefing.com/solana-music-launch-spotify-challenger/>

¹³⁹ Society of Environmental Journalists (citing New York Times analysis). <https://www.sej.org/headlines/thousands-us-government-web-pages-have-been-taken-down-friday>

¹⁴⁰ Freie Universität Berlin, Earth Sciences Library.

<https://www.geo.fu-berlin.de/en/bibliotheken/Aktuelles/Verschwindende-Daten---Trump-Administration-entfernt-Klimainformationen-von-Regierungswebsites.html>

¹⁴¹ WITNESS (written submission to the UN Special Rapporteur on freedom of expression, via OHCHR, citing Syrian Archive).

<https://www.ohchr.org/Documents/Issues/Opinion/ContentRegulation/Witness.pdf>

and the lineage is chain state other contracts can read¹⁴² — and the collection-mint contract already splits proceeds between artist, marketplace, and operator, so the pattern exists to extend¹⁴³. The pain justifies the build: clearing one sample costs \$500–\$50,000 and takes months, and the US Mechanical Licensing Collective sits on an estimated ~\$397 million in unmatched royalties because attribution metadata fails¹⁴⁴. The build waits for month three because it needs an audience — the stunts manufacture exactly that — with XTRATA FM as the venue and a 100-remix on-chain relay as the proving demo. The honest framing stands: attribution rails, not legal clearance.

5.4 Prioritization Table

5.4.1 Table: use case × attention potential × pain severity × build effort × time-to-launch × recommended order

Order	Use case	Attention potential	Pain severity	Build effort	Time-to-launch
0	Developer-agent surface fixes (npm, llms.txt, MCP, skill, bounty config)	Low direct — gates everything	Gating: converts all future press	Days	Week 0–4
1	UC1 The album that outlives the label	Very high — Nina goes dark 2026-07-22 ¹²⁹	Very high	Low — tooling live	This week
2	UC12 Monthly rescue cadence	Very high — meta-headline per stunt	High (attention is the pain)	Low — marketing program	Month 1, ongoing
3	UC5 Civic data rescue	High	High	Low — rights-free federal data	Month 2
4	UC4 War-crime evidence vault	High	Very high — 120–150k videos erased ¹⁴¹	Medium — NGO partner + abuse policy	Month 3

¹⁴² Xtrata (stxtrata/xtrata). <https://github.com/stxtrata/xtrata/blob/main-staging/xtrata-1.0/docs/recursive-inscriptions.md>

¹⁴³ Xtrata (stxtrata/xtrata). <https://github.com/stxtrata/xtrata/blob/main-staging/xtrata-1.0/docs/contract-inventory.md>

¹⁴⁴ Chartlex. <https://www.chartlex.com/blog/business/sample-clearance-guide-musicians-2026>

¹⁴⁵ Interspace Music. <https://interspacemusic.com/blog/the-mlc-has-paid-out-billion-the-money-stuck-in-its-black-box-gets-split-by-market-share/>

Order	Use case	Attention potential	Pain severity	Build effort	Time-to-launch
5	UC3 XTRATA FM embed-anywhere push	High	Medium	Low — radio already live	Month 3–4
6	UC2 Remix royalty economy	High	Very high — ~\$397M unmatched royalties ¹⁴⁵	Medium-high — one new contract	Month 3–9
7	UC7 Certified-human registry	Medium — EU AI Act rule hits 2026-08-02 ¹⁴⁶	High	Low — wizard flow + badge	Month 4–6
8	UC8 Agents that publish forever	Medium	Medium	Low — once 5.1 lands	Month 4–6
9	UC9 The arcade that cannot be delisted	High — 1.3M-signature movement ¹⁴⁷	High	Medium — needs one indie studio	Month 6–9
10	UC6 Self-correcting scientific record	Medium	High	Medium — journal or funder pilot	Month 6–12
11	UC10 Composable game assets	Medium	Medium	High — engine loader needed	Month 9–12
12	UC11 The digital will	Low near-term	High	High — trigger oracle + legal	Year 2 moonshot

The ordering rule is attention per unit of engineering. Everything before row 6 requires zero new contracts — the top five use-case rows all run on code already on mainnet, so the binding constraint is nerve and rights-clearance, not engineering capacity. Pain severity and attention do not always travel together: UC6 and UC11 sit on severe pain but demand partners and layers that do not yet exist, so they queue behind rows that convert press into users first. The table also encodes Chapter 4’s discipline: every row sells relationships, ownership, or sponsorship — never bulk

¹⁴⁶ Truescreen. <https://truescreen.io/articles/c2pa-standard-history-limitations/>

¹⁴⁷ GameDeveloper (Reuters). <https://www.gamedeveloper.com/business/french-consumer-group-sues-ubisoft-over-shutdown-of-the-crew>

storage. Execute only rows 0–3 and Xtrata enters the autumn with press, claiming users, and an agent-readable surface — the assets the heavy builds need to land.

What happens next is dated. Nina Protocol’s servers go dark tomorrow, and every week of silence cedes the “permanent home for music” narrative to whoever ships first — a narrative whose last two claimants died with their media off-chain. The machine is built; the audience is the deliverable. Inscribe the first artifact this week.

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